Sreenath Paleri

Website: sreenathpaleri.github.io *Email:* paleri@tamu.edu

Education

University of Wisconsin Madison **Sep 2018-May 2023**
Ph.D. Atmospheric and Oceanic Sciences

Dissertation: Role of Heterogeneous Ecosystems in Modulating Surface-Atmosphere Transport
Advisor: Dr. Ankur Desai, Ecometeorology Lab

Indian Institute of Technology Madras, Chennai, India **2008-2013**
B. Tech. in Civil Engineering and M. Tech. in Applied Mechanics (Specialization in Fluid Mechanics)

Master's Thesis: Interaction of Inertial Particles with Line Plumes in Natural Convection
Advisor: Dr. Baburaj Puthenveetil, Laser Velocimetry and Fluorescence Lab

Research Interests

Simulations and simple models of energy, water and mass fluxes in the turbulent atmospheric boundary layer. Large eddy simulations, tower mounted and airborne eddy covariance, wavelet analysis of land-atmosphere interactions, scaling laws for the heterogeneously heated atmospheric boundary layer

Appointments and Positions

Department of Atmospheric Sciences, **Aug 2025 - Present**
Texas A&M University College Station, TX, USA
Assistant Professor

National Ecological Observatory Network (NEON) program, **Mar 2025 - Aug 2025**
Battelle Memorial Institute Boulder, CO, USA
Environmental Scientist II

Cooperative Institute for Severe and High-Impact Weather Research and Operations,
University of Oklahoma **July 2023 - Mar 2025**
Postdoctoral Research Associate Boulder, CO, USA

NOAA Air Resources Laboratory, **July 2023 - Mar 2025**
Atmospheric Turbulence and Diffusion Division Boulder, CO, USA
Postdoctoral Affiliate

Department of Atmospheric and Oceanic Sciences, **Sep 2018 - May 2023**
UW Madison
Research Assistant, with Prof. Ankur Desai. Madison, Wisconsin USA

Institute of Hydrology and Meteorology, **Feb - Apr 2022**
Technische Universität Dresden
Visiting Researcher, hosted by Prof. Dr. Matthias Mauder Dresden, Germany

Centre for Atmospheric and Oceanic Sciences, **Feb 2016 - Jul 2018**
Indian Institute of Science
Project Scientist, with Prof. Bhat G. S. Bangalore, India

Department of Fluid Mechanics, IIT Madras **Jan 2012-Mar 2012**
Research Intern, with Dr. Patnaik P. Chennai, India

PUBLICATIONS

Book chapter

3 - Here, there, and everywhere: Spatial patterns and scales. In A. L. Hiscox (Ed.), Conceptual Boundary Layer Meteorology (pp. 37–58)., Elsevier **2023**
Paleri S.,*Butterworth, B., Desai, A.R.*
Academic Press, DOI:10.1016/B978-0-12-817092-2.00009-6

Peer reviewed

Impact of Surface Heterogeneity Induced Secondary Circulations on the Atmospheric Boundary Layer **2025**
Paleri S.,*Wanner L., Sühring M., Desai A., Butterworth B., Mauder M., Metzger S.*
Boundary Layer Meteorology, DOI:10.1007/s10546-024-00893-7

Space - scale resolved surface-atmospheric fluxes across a heterogeneous midlatitude forested landscape **2022**
Paleri S., *Desai A., Metzger M., Durden D., Butterworth B., Mauder M., Kohnert K., Serafimovich A.*
Journal of Geophysical Research - Atmospheres, DOI:10.1029/2022JD037138

Scaling Land-Atmosphere Interactions: Special or Fundamental? **2022**
Ankur R. Desai, ***Sreenath Paleri,*** *James Mineau, Hawwa Kadum, Luise Wanner, Matthias Mauder, Brian J. Butterworth, David J. Durden, Stefan Metzger*
Journal of Geophysical Research: Biogeosciences, DOI:10.1029/2022JG007097

Towards Energy-Balance Closure With a Model of Dispersive Heat Fluxes **2024**
Wanner L., Jung M., ***Paleri S.****, Kadum H., Desai A., Mauder M.*
Boundary Layer Meteorology, DOI:10.1007/s10546-024-00868-8

Characterizing Energy Balance Closure over a Heterogeneous Ecosystem Using Multi-Tower Eddy Covariance **2024**
Butterworth B., Desai A., Durden D., Kadum H., LaLuzerne D., Mauder M., Metzger S., ***Paleri S.****, Wanner L.*
Frontiers in Earth Science, DOI: 10.3389/feart.2023.1251138

Airborne Measurements of Scale-Dependent Latent Heat Flux Impacted by Water Vapor and Vertical Velocity over Heterogeneous Land Surfaces During the CHEESEHEAD19 Campaign **2024**
Lin G., Wang Z., Chu Y., Ziegler C., Hu X., Xue M., Geerts B., ***Paleri S.****, Desai A., Yang K., Deng M., DeGraw J.* Journal of Geophysical Research - Atmospheres, DOI:10.1029/2023JD039586

Drivers of decadal carbon fluxes across temperate ecosystems **2022**
Desai A.R., Murphy B., Wiesner S., Thom J. E. , Butterworth B. J., Koupaei-Abyazani N., Muttaqin A. S., ***Paleri S.****, Talib A., Turner J., Mineau J., Merrelli A., Stoy P. C., Davis K.*
Journal of Geophysical Research - Biogeosciences, DOI:10.1029/2022JG007014

Novel approach to observing system simulation experiments improves information gain of surface-atmosphere field measurements **2021**
S.Metzger, D. Durden, ***S. Paleri****, M. Sühring, B. Butterworth, A. Desai, C. Florian, M. Mauder, D. Plummer, E. Sigel, Z. Wang, L. Wanner, K. Xu*
Atmospheric Measurement Techniques, DOI:10.5194/amt-14-6929-2021

Multi-sensor approach for high space and time resolution land surface temperature
Ankur R. Desai, Anam M. Khan, Ting Zheng, **Sreenath Paleri***, Brian Butterworth, Temple R. Lee, Joshua B. Fisher, Glynn Hulley, Tania Kleynhans, Aaron Gerace, Philip A. Townsend, Paul Stoy, Stefan Metzger* **2021**
Earth and Space Science, DOI:10.1029/2021EA001842

Ecosystem-atmosphere exchanges of carbon dioxide, water vapour and energy in India: a synthesis of insights from eddy covariance measurements
Burman et al. **2025**
Agricultural and Forest Meteorology, DOI:10.1016/j.agrformet.2025.110730

Connecting Land–Atmosphere Interactions to Surface Heterogeneity in CHEESEHEAD19
Butterworth et al. **2020**
Bulletin of the American Meteorological Society, DOI:10.1175/BAMS-D-19-0346.1

Spatial and temporal variability in energy and water vapour fluxes observed at seven sites on the Indian subcontinent during 2017 **2020**
G. S. Bhat, R. Morrison, C. M. Taylor, B. K. Bhattacharya, ***S. Paleri*** *, D. Desai, J. G. Evans, S. Pattnaik, M. Sekhar, R. Nigam, A. Sattar, S. S. Angadi, D. Kacha, A. Patidar, S. N. Tripathi, K. V. M. Krishnan, A. Sisodiya*
Quarterly Journal of the Royal Meteorological Society, DOI:10.1002/qj.3688

Interaction of convective organization with monsoon precipitation, atmosphere, surface and sea: The 2016 INCOMPASS field campaign in India **2020**
A. G. Turner, G. S. Bhat et al.
Quarterly Journal of the Royal Meteorological Society, DOI:10.1002/qj.3633

Under Review
Where Has All the Energy Gone? Quantifying Advective Energy Fluxes with a Dense Tower Network
Mather E., Desai A., Blakely B., Butterworth B.,Metzger S., ***Paleri S.***

In Preparation

Evaluation of the PALM model system in simulating the observed atmospheric boundary layer diurnal evolution over a heterogeneous, forested landscape during the CHEESEHEAD19 field campaign
Paleri S.,*Wanner L., Sühring M., Desai A., Butterworth B., Mauder M.*

A generalized bulk parameterization approach for land-atmosphere heat fluxes
Paleri S.,*Meyers T., Lee T., Heuer M.*

Observations of surface layer turbulence, surface energy, and water budget in complex terrain during the SPLASH field campaign
Paleri S.,*Meyers T., Heuer M.*

Professional Service

- Editorial Board Member, Boundary Layer Meteorology, **Feb. 23 - Present**
- AMS Committee on Measurements, **Jan 24 - Present**
- Review panellist, NASA Decadal survey incubation program **Mar 2025**
- Session co-chair, AMS Symposium on Measurements and Instrumentation **Jan 2025, 2026**
- Session co-chair, AMS 21st Conference on Mountain Meteorology **June 2024**
- Session co-chair, AMS 34th Agricultural and Forest Meteorology conference **June 2021**
- Organizer, FLUXNET Community workshop : Bridging the Gap: Flux Data Meets Land Surface Models, Berkeley, CA **Aug 2025**
- FLUXNET Early Career Network, **July 23 - Present**

Invited Presentations

- **Measuring and Modeling Multi-Scale Surface Atmosphere Exchanges** **Feb. 2024**
  Atmospheric Sciences Seminar,
  Department of Atmospheric Science, Colorado State University
- **Interactions between land surface heterogeneity and lower atmospheric response** **Mar. 2022**
  Atmospheres and Oceans seminar,
  Department of Earth and Planetary Sciences, Johns Hopkins
- **Using high resolution observations and realistic LES to study the interactions between land surface heterogeneity and lower atmospheric response** **Nov. 2020**
  Online, Coupling of Land and Atmospheric Subgrid Parameterizations (CLASP) Project Meeting
- **Interactions between land surface heterogeneity and scale transport in the Atmospheric Boundary Layer** **Oct. 2019**
  RUBISCO-Ameriflux Workshop, Lawrence Berkeley National Lab, Berkeley, CA.

Oral Presentations

- **Using Large Eddy Simulations to Study the Interactions Between Land Surface Heterogeneity and Lower Atmospheric Response During the CHEESEHEAD19 Field Campaign.** **Dec. 2022**
  American Geophysical Union, Fall Meeting
- **Diagnosing the Effects of Surface Heterogeneity Induced Secondary Circulations from Large Eddy Simulations of Diurnal Cycles During the CHEESEHEAD19 Field Campaign** **Dec. 2021**
  American Geophysical Union, Fall Meeting
- **Scale resolved, sub-grid surface fluxes across a heterogeneous mid-latitude forested landscape** **June 2021**
  Online, AMS 34th Conference on Agricultural and Forest Meteorology
- **Representativeness of surface heterogeneity induced secondary atmospheric circulations in Large Eddy Simulations** **Dec. 2020**
  Oral Session, Online, American Geophysical Union, Fall Meeting
- **Scale resolved, sub-grid surface fluxes across a heterogeneous mid-latitude forested landscape** **June 2020**
  Online, CHEESEHEAD mini conference, UW Madison
- **Scale resolved flux contributions to surface atmospheric interactions across a mid-latitude, heterogeneous landscape from airborne measurements** **Nov. 2020**
  UW AOSS Department Seminar, Madison, Wisconsin
- **Mesoscale flux contributions to surface-atmosphere interactions across a heterogeneous mid-latitude landscape** **Dec. 2019**
  Oral Session, American Geophysical Union, Fall Meeting

Professional Experience

Cognizant Technology Solutions, **July 2013-Feb2016**
Associate, Business Analytics Chennai, India

- Developed statistical models using **logistic regression** and **Principal Component Analysis** on **big data** to predict consumer behavior for multiple marketing analytics projects

Cochin Port Trust, Government of India, **May -July 2011**
Industrial Intern Cochin, India

Community Service and Outreach

- Journal Reviewer, Boundary Layer Meteorology, Geophysical Research Letters, Journal of Atmospheric Sciences, Agricultural and Forest Meteorology, Journal of Applied Meteorology and Climatology
- Student coach , Meteorology Science Olympiad, **2022-23**
  Hamilton Middle School, Madison WI
- Volunteer, Wisconsin Science Festival, **2022-23**
- Judge, NASA GLOBE Midwest Earth System Science Symposium **May 2021**
- Student Mentor, Atmospheric and Oceanic Sciences Undergraduate Mentorship Program. University of Wisconsin-Madison **2019-Present**

Teaching and Mentoring

- Mentor, Atmospheric and Oceanic Sciences Undergraduate Mentorship Program, University of Wisconsin-Madison **2020, 2021, 2022, 2023**
  Undergraduates mentored: Tanner Blanke, Derek Hsu, Tyler Varana, Benjamin Zumwalt
- Teaching Assistant, Introduction to Weather and Climate, Departmnet of Atmospheric and Oceanic Sciences, UW Madison **August -December 2018**

- Teaching Assistant, Atmosphere-Ocean Fluid Dynamics, Centre for Atmospheric and Oceanic Sciences (CAOS), IISc Bangalore. **January-March 2018**
- Teaching Assistant, Observational Techniques, Centre for Atmospheric and Oceanic Sciences (CAOS), IISc Bangalore. **August -December 2017**
- Teaching Assistant, Engineering Mechanics, Department of Applied Mechanics, IIT Madras **August -December 2012**

Workshops and Professional Development

- Bridging the Gap: Flux Data Meets Land Surface Models, Berkeley, CA **Aug 2025**
- Remote Sensing and Fluxes Upscaling for Real-world Impact, Lawrence Berkeley National Lab, Berkeley, CA **Sep 2024**
- Field Experiment on Submesoscale Spatio-Temporal Variability in Lindenberg (FESSTVal) Summer School Scharmützelsee (Brandenburg), near Berlin, MOL-RAO Germany (postponed from 2020 due to COVID-19 and finally cancelled) **2022**
- Unlearning Racism in Geosciences (URGE) **Feb.-May 2021**
  University of Wisconsin-Madison
- Reducing Uncertainties in Biogeochemical Interactions through Synthesis and Computation (RUBISCO) Workshop **Oct. 2019**
  University of California, Berkely, CA
- Teaching in Science and Engineering **Sept.-Dec. 2018**
  University of Wisconsin-Madison
- Summer School and Discussion Meeting on Buoyancy Driven Flows **2018**
  International Center for Theoretical Sciences , Bangalore, India